

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE VIII

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG
MARÇO/2026

RAMON LOPES DE QUEIROZ

ATIVIDADE VIII

Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Cálculo Integral e Diferencial do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

Professor: Daniel Oliveira

MONTES CLAROS – MG

MARÇO/2026

Exercício 8.

a. $f(x) = \frac{x^2 + x - 6}{x + 3}$

$x = -3$

↳ Denominator zero, ou seja, $f(-3)$ não existe.

b. $g(x) = \begin{cases} \frac{1}{x+2}, & x \neq -2 \\ 0, & x = -2 \end{cases}$ $\lim_{x \rightarrow -2^-} \frac{1}{x+2} = -\infty$ $\lim_{x \rightarrow -2^+} \frac{1}{x+2} = +\infty$

$\lim_{x \rightarrow -2} g(x) = g(-2)$ → O limite não existe, pois seus limites laterais são infinitos.

c. $H(x) = \begin{cases} 1+x, & x \leq -2 \\ 2-x, & -2 < x \leq 2 \\ 2x-1, & x > 2 \end{cases}$

$x = -2$

$H(-2) = 1 - 2 = -1$

$\lim_{x \rightarrow -2^-} (1+x) = -1$

$\lim_{x \rightarrow -2^+} H(x)$ não existe (laterais diferentes)

$\lim_{x \rightarrow 2^-} (2-x) = 0$ $(-1 \neq 0)$

d. $F(x) = \begin{cases} \sqrt{-x}, & x < 0 \\ \sqrt[3]{x+1}, & x \geq 0 \end{cases}$

$F(0) = \sqrt[3]{0+1} = 1$

$\lim_{x \rightarrow 0^-} \sqrt{-x} = 0$

limites laterais são diferentes, ou seja, limite não existe.

$\lim_{x \rightarrow 0^+} \sqrt[3]{x+1} = 1$

$$e. f(x) = \begin{cases} 1+x, & x \neq -2 \\ 3, & x = -2 \end{cases}$$

$$f(-2) = 3 \quad \lim_{x \rightarrow -2} (1+x) = 0$$

$$\lim_{x \rightarrow -2} f(x) \neq f(-2)$$

Apuntes de la lección
b = -2.

$$2-11. \lim_{t \rightarrow \infty} \frac{2t+1}{5t-2} = \frac{\frac{2t}{t} + \frac{1}{t}}{\frac{5t}{t} - \frac{2}{t}} = \frac{2+0}{5-0} = \frac{2}{5}$$

$$24- \lim_{x \rightarrow \infty} \frac{5x^3 - 12x + 7}{4x^2 - 1} = \frac{\frac{5x^3}{x^2} - \frac{12x}{x^2} + \frac{7}{x^2}}{\frac{4x^2}{x^2} - \frac{1}{x^2}} = \frac{5x - \frac{12}{x} + \frac{7}{x^2}}{4 - \frac{1}{x^2}} = \frac{-\infty - 0 + 0}{4 - 0} = -\infty$$

$$29- \lim_{w \rightarrow -\infty} \frac{\sqrt{w^3 - 2w + 3}}{w + 5} = \frac{\sqrt{w^3(1 - \frac{2}{w} + \frac{3}{w^3})}}{w(1 + \frac{5}{w})} = \frac{-w\sqrt{1 - \frac{2}{w} + \frac{3}{w^3}}}{w(1 + \frac{5}{w})} = -1 \cdot \frac{\sqrt{1 - 0 + 0}}{(1 + 0)}$$

$$= \frac{-1\sqrt{1-0+0}}{1+0} = \frac{-1 \cdot 1}{1} = -1$$

$$31- \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) = x\sqrt{1} - x = 0$$

$$33- \lim_{n \rightarrow \infty} (\sqrt{3n^2 + n} - 2n) = \frac{(\sqrt{3n^2 + n} - 2n)(\sqrt{3n^2 + n} + 2n)}{(\sqrt{3n^2 + n} + 2n)} = \frac{(\sqrt{3n^2 + n})^2 - (2n)^2}{\sqrt{3n^2 + n} + 2n} = \frac{(\sqrt{3n^2 + n})^2 - 2n^2}{\sqrt{3n^2 + n} + 2n}$$

$$= \frac{3n^2 + n - 4n^2}{\sqrt{3n^2 + n} + 2n} = \frac{-n^2 + n}{\sqrt{3n^2 + n} + 2n} = \frac{-\frac{n^2}{n} + \frac{n}{n}}{\sqrt{3 + \frac{1}{n}} + 2} = \frac{-n + 1}{\sqrt{3 + \frac{1}{n}} + 2} = \frac{-\infty}{\sqrt{3} + 2} = -\infty$$

$$36- \lim_{x \rightarrow \infty} \frac{\sqrt{x}(\sqrt{x+1} - \sqrt{x})}{\sqrt{x+1}} = \frac{\sqrt{x} \left(\frac{1 + \sqrt{x+1} - \sqrt{x}}{x} \right)}{\sqrt{x+1}} = \frac{\sqrt{x} \cdot \frac{1 + \sqrt{x+1} - \sqrt{x}}{x}}{\sqrt{x+1}} = \frac{\sqrt{x} \cdot \frac{1 + \sqrt{x+1} - \sqrt{x}}{x}}{\sqrt{x+1}}$$

$$\frac{\sqrt{1 + \frac{1 + \sqrt{x+1} - \sqrt{x}}{x}}}{\sqrt{1 + \frac{1}{x}}} = \frac{\sqrt{1 + 0}}{\sqrt{1 + 0}} = 1$$

45. $f(x) = \frac{2x}{6x^2 + 11x - 10}$

$\lim_{x \rightarrow \infty} \frac{2x}{6x^2 + 11x - 10} = \frac{2 \cdot 0}{6 \cdot \frac{1}{2} + \frac{11}{2} - \frac{10}{2}} = \frac{2 \cdot 0}{9}$

$6x^2 + 11x - 10 = 0$

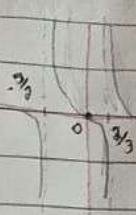
$\Delta = 121 - 4 \cdot 6 \cdot (-10)$

$\Delta = 361$

$x = \frac{-11 \pm 19}{12} = x_1 = \frac{2}{3}$
 $x_2 = -\frac{5}{2}$

$AV = \frac{2}{3} \cdot -\frac{5}{2}$
 $AV = -\frac{5}{3}$

$AM = 0$ (Cena 2)



47. $\frac{4x^2}{\sqrt{x^2 - 2}}$

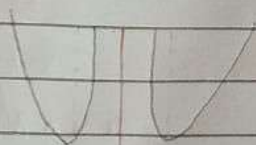
$\lim_{x \rightarrow \pm\infty} \frac{4x^2}{\sqrt{x^2 - 2}} = \lim_{x \rightarrow \pm\infty} = +\infty$

$\lim_{x \rightarrow -\infty} = +\infty$

$\sqrt{x^2 - 2} > 0 \Rightarrow x^2 > 2$

$(-\infty, -\sqrt{2}) \cup (\sqrt{2}, +\infty)$

$AV = \sqrt{2} \cdot -\sqrt{2}$



-2 2

54. $2xy^2 + 4y^2 = 3x - 0$

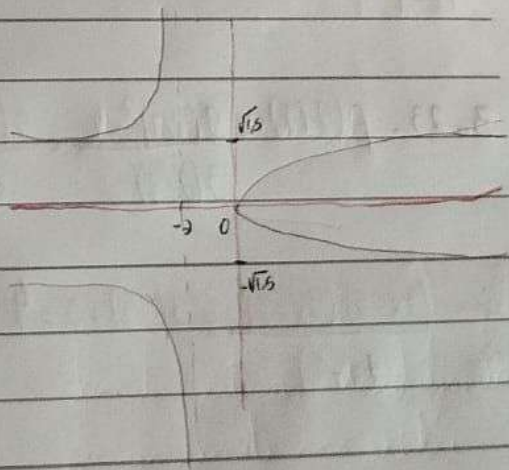
$y^2(2x + 4) = 3x$

$y^2 = \frac{3x}{2x + 4}$

$y = \pm \sqrt{\frac{3x}{2x + 4}}$

$\lim_{x \rightarrow \pm\infty} \pm \sqrt{\frac{3x}{2x + 4}} = \pm \sqrt{\frac{3}{2}}$

$y = \sqrt{1.5}$
 $y = -\sqrt{1.5}$
 $AM = 0$



$2x + 4$

$x = -2$

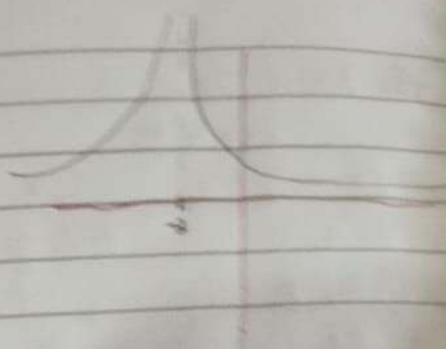
AV

$$56. x^2(x+4) - x^2(x+4) - 6 = 0$$

$$y(x^2(x+4) - x^2(x+4) - 6) = 0 \quad \lim_{x \rightarrow \infty} \frac{x^2(x+4) - x^2(x+4) - 6}{x^2(x+4)} = \frac{-6}{x^3} = 0$$

$$\frac{-6}{x^3} = 0$$

$$x = -2$$



$$57. f(x) = \frac{x}{x-1}$$

$$\frac{1}{x-1}$$

$$\varepsilon > 0, N = \frac{1}{\varepsilon} + 1$$

$$|f(x) - 1| < \varepsilon$$

$$x-1 > \frac{1}{\varepsilon}$$

$$\frac{1}{x-1} < \varepsilon, x > 1$$

$$\left| \frac{x}{x-1} - 1 \right| < \varepsilon$$

$$\frac{1}{x-1} < \varepsilon$$

$$x > N, |f(x) - 1| < \varepsilon$$

$$\left| \frac{x - (x-1)}{x-1} \right| < \varepsilon$$

$$x > \frac{1}{\varepsilon} + 1$$

$$\varepsilon x = N = \frac{1}{\varepsilon} + 1 = 101$$

$$\left| \frac{1}{x-1} \right| < \varepsilon$$

$$3.23. f(2/3) = \frac{9(2/3)^2 - 4}{3(2/3) - 2} = \frac{9(4/9) - 4}{2 - 2} = \frac{0}{0}$$

$$\lim_{x \rightarrow 2/3} \frac{(3x-2)(3x+2)}{(3x-2)} = 3\left(\frac{2}{3}\right) + 2 = 4$$

$$\begin{cases} \frac{9x^2-4}{3x-2} & \text{if } x \neq 2/3 \\ 4 & \text{if } x = 2/3 \end{cases}$$

$$f(x) = 3x + 2, \text{ for all } x \in \mathbb{R}$$

$$29. f(x) = \begin{cases} |x-3| & \text{se } x \neq 3 \\ 2 & \text{se } x = 3 \end{cases}, a = 3$$

$$f(x) = \begin{cases} |x-3| & \text{se } x \neq 3 \\ 0 & \text{se } x = 3 \end{cases}$$

$$f(3) = 2$$

Descontinuidad removible.

$$\lim_{x \rightarrow 3} |x-3| = |3-3| = 0$$

$$32. f(x) = \frac{\sqrt[3]{x+1} - 1}{x}, a = 0$$

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{x+1} - 1}{x}$$

$$u = \sqrt[3]{x+1}$$

$$u^3 = x+1$$

$$x = u^3 - 1$$

$$f(0) = \frac{\sqrt[3]{0+1} - 1}{0} = \frac{0}{0}$$

$$\lim_{u \rightarrow 1} \frac{u-1}{u^3-1} = \frac{u-1}{(u-1)(u^2+u+1)}$$

descontinua en $a=0$

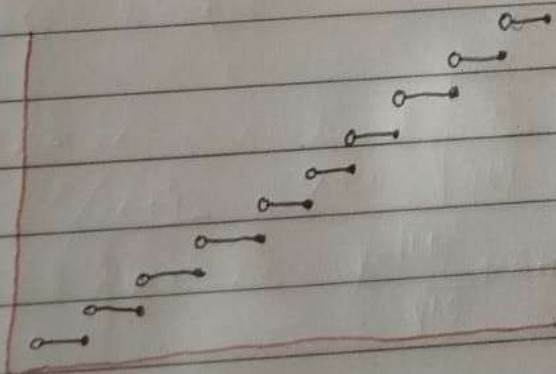
$$\lim_{u \rightarrow 1} \frac{1}{u^2+u+1} = \frac{1}{3} \quad \text{descont. removible}$$

$$f(x) = \begin{cases} \sqrt[3]{x+1} - 1/x & \text{se } x \neq 0 \\ 1/3 & \text{se } x = 0 \end{cases}$$

$$44. f(x) = 0,22 + 0,17 \cdot \left\lceil \frac{x-30}{30} \right\rceil, \text{ para } x > 30$$

límite izquierdo = grado anterior
límite derecho = nuevo grado

$$P = x \in \{30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330\}$$



$$46. f(x) = \frac{\sqrt{2+\sqrt[3]{x}} - 2}{x-8}$$

$$f(8) = \frac{\sqrt{2+\sqrt[3]{8}} - 2}{8-8} = \frac{0}{0}$$

$$\lim_{x \rightarrow 8} \frac{\sqrt{2+\sqrt[3]{x}} - 2}{x-8} \cdot \frac{\sqrt{2+\sqrt[3]{x}} + 2}{\sqrt{2+\sqrt[3]{x}} + 2} = \frac{(\sqrt{2+\sqrt[3]{x}})^2 - 4}{(x-8)(\sqrt{2+\sqrt[3]{x}} + 2)} = \frac{\sqrt[3]{x} - 2}{(x-8)(\sqrt{2+\sqrt[3]{x}} + 2)}$$

$$\frac{1}{(\sqrt[3]{8^2} + 2\sqrt[3]{8} + 4)(\sqrt{2+2} + 2)} = \frac{1}{12 \cdot 4} = \frac{1}{48}$$

$$f(x) = \begin{cases} \frac{\sqrt{2+\sqrt[3]{x}} - 2}{x-8} & \text{se } x \neq 8 \\ 1/48 & \text{se } x = 8 \end{cases}$$

$$50. f(x) = x$$

$$h(0) = 0$$

contínua em 0.

$$g(x) = \begin{cases} 1 & \text{se } x \neq 0 \\ 0 & \text{se } x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} h(x) = 0$$

$$x \rightarrow 0$$

$$x = \lim$$

$$(f \cdot g)(x) = h(x)$$

$$\text{se } x \neq 0 \rightarrow h(x) = x \cdot 1 = x$$

$$\text{se } x = 0 \rightarrow h(x) = 0 \cdot 0 = 0$$

$$h(x) = x$$